

# MICRODUCK PHYSICAL AI MASTERCLASS

PHASE 3 HANDOUT

Module 3: The Dog Trainer: Gym Environments, Reward Engineering, PPO & Curriculum Learning

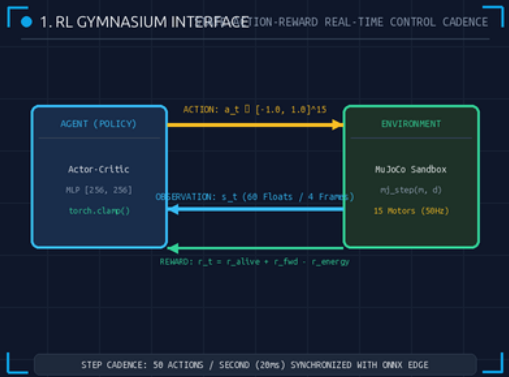
RL & PPO

ALGORITHM: PPO (Clipped)

OBSERVATION: 60 Floats (4 Frames)

ACTION: 15 Clamped [-1, 1]

FREQUENCY: 50 Hz Control Loop



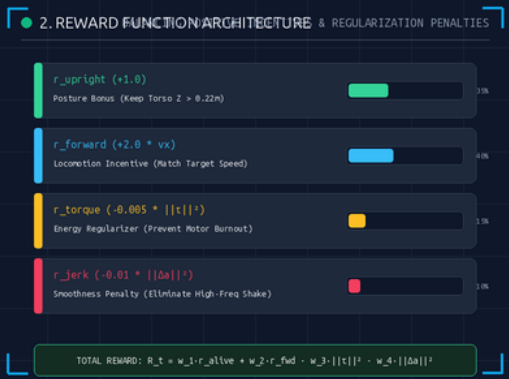
## 1. Gymnasium Environment & State Loop

OBSERVATION & ACTION CADENCE

**The RL Paradigm:** The policy observes a sliding history tensor  $\mathbb{R}^{60}$ , infers 15 normalized motor velocity actions  $\mathbb{a}_t \in [-1.0, 1.0]$ , and receives a scalar reward.

**The 12-Year-Old Analogy:** When you train a puppy to fetch, you don't pick up its paws. You show it the ball, watch what it does, and toss a biscuit when it brings it back.

- **Observation Space:** Box(-inf, inf, shape=(60,), float32) representing 4 frames  $\times$  15 sensors.
- **Action Space:** Box(-1.0, 1.0, shape=(15,), float32) target joint velocities.
- **50Hz Execution:** Strict 20ms step cadence ensuring zero sim-to-real timing discrepancy.



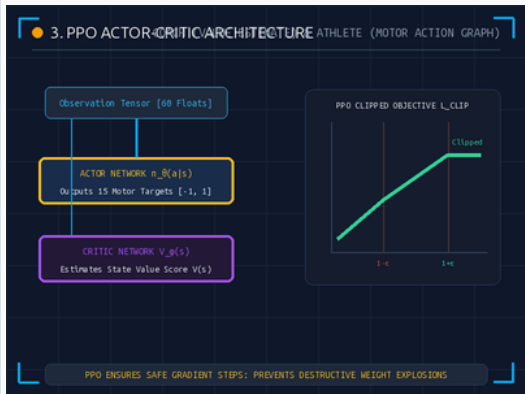
## 2. Reward Engineering & Penalties

BALANCING PROGRESS VS ENERGY BURN

**Mathematical Incentive Design:**  $R_t = w_{\text{up}} r_{\text{up}} + w_{\text{fwd}} r_{\text{fwd}} - w_{\text{tau}} \|\tau\|^2 - w_{\text{j}} \|\Delta a\|^2$ . The agent is penalized for high torque and action jerk.

**The 12-Year-Old Analogy:** If you only reward speed, the duck will throw itself down the stairs to cross the line. You must penalize falling and shaking.

- **Upright Bonus:**  $r_{\text{up}} = +1.0$  if torso  $z > 0.22\text{m}$  (immediate termination if  $z < 0.14\text{m}$ ).
- **Velocity Tracking:**  $r_{\text{fwd}} = \exp(-2.0 \cdot (v_x - v_{\text{target}})^2)$  matching target forward pace.
- **Torque Regularizer:**  $-0.005 \cdot \|\tau\|^2$  prevents motor overheating and gear wear.



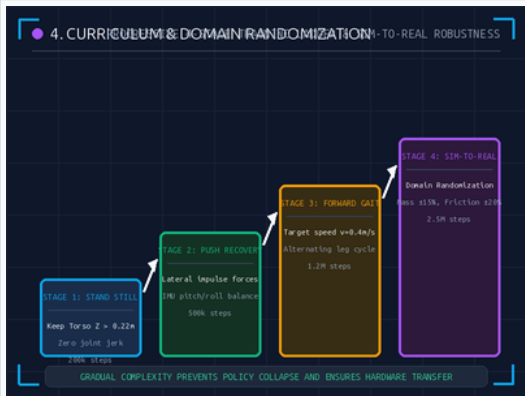
### 3. PPO: Actor-Critic Architecture

COACH VS ATHLETE &  
CLIPPED OBJECTIVE

**Proximal Policy Optimization:** The Actor network  $\pi_{\theta}(a|s)$  drives the joints, while the Critic network  $V_{\phi}(s)$  evaluates performance. The clipped surrogate objective prevents destructive weight updates.

**The 12-Year-Old Analogy:** The Gymnast (Actor) performs a trick; the Coach (Critic) scores it. PPO stops the Gymnast from making crazy changes that erase past lessons.

- **Clipped Envelope:**  $L^{CLIP}(\theta) = \hat{E}_t [\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon) \hat{A}_t)]$ , with  $\epsilon=0.2$ .
- **Advantage Estimation:** GAE  $(\lambda=0.95)$  computes temporal difference score  $\hat{A}_t$  to guide policy gradient.
- **Pruned Edge Export:** Critic is discarded post-training; only lightweight Actor is exported to ONNX.



### 4. Curriculum & Sim-to-Real Transfer

4-STAGE LADDER &  
DOMAIN RANDOMIZATION

**Progressive Training:** Locomotion is unlocked across 4 stages (Stand  $\rightarrow$  Push Recovery  $\rightarrow$  Gait  $\rightarrow$  Domain Randomization) to avoid catastrophic policy collapse.

**The 12-Year-Old Analogy:** First teach a child to stand on two feet. Then teach balance against gentle shoves. Then take a step. Then run on wet grass.

- **Stage Progression:** Stage 1 (200k steps)  $\rightarrow$  Stage 2 (500k steps)  $\rightarrow$  Stage 3 (1.2M steps)  $\rightarrow$  Stage 4 (2.5M steps).
- **Domain Randomization:** Perturbs link mass  $(\pm 15\%)$ , ground friction  $(\pm 20\%)$ , and latency  $(1 \dots 5 \text{ms})$ .
- **Hardware Safety Clamp:** `torch.clamp(a, -1.0, 1.0)` baked in silicon guarantees 0% gear stripping.

## ■ PHYSICAL AI COMPLETE MASTERCLASS CERTIFICATION

- **Hardware (Phase 1):** RK3566 onboard compute, 6-DOF IMU, 15 actuators, and battery constraints.
- **Simulation (Phase 2):** MuJoCo kinematic trees, collision geoms, autolimits, and 50Hz integration.
- **Control (Phase 3):** PPO reinforcement learning, clamped action graphs, and sim-to-real transfer.

# ■ PHASE 3 KNOWLEDGE CHECK

ASSESSMENT

Test your understanding of Reinforcement Learning, PPO, and Sim-to-Real transfer.

5 Questions

## ■ Q1: Temporal Observation Dimensions

MULTIPLE CHOICE

How many float values make up the Microduck's 4-frame observation tensor?

- A. 15 floats
- B. 60 floats (4 historical frames × 15 sensors)
- C. 1,000 floats

## ■ Q2: Torque Penalty in Reward Design

MULTIPLE CHOICE

Why must reward functions include a negative torque penalty term ( $-||\tau||^2$ )?

- A. To eliminate high-frequency motor shaking and conserve battery energy.
- B. Because torque does not exist in real robotics.
- C. To force the robot to walk backwards.

## ■ Q3: Actor-Critic Roles

MULTIPLE CHOICE

In Actor-Critic architectures like PPO, what is the specific role of the Critic?

- A. It directly sends PWM electrical commands to the motors.
- B. It acts as the Coach, predicting expected future discounted returns  $V(s)$ .
- C. It regulates the cooling fans on the Rockchip CPU.

## ■ Q4: Silicon Safety Clamping

MULTIPLE CHOICE

Why is `torch.clamp(a, -1.0, 1.0)` baked directly into the ONNX graph?

- A. To mathematically guarantee motor targets never exceed physical limits, preventing gear stripping.
- B. To reduce the file size of the neural network.
- C. To activate the RGB camera lens.

## ■ Q5: Sim-to-Real Transfer

MULTIPLE CHOICE

What is Domain Randomization during curriculum training?

- A. Playing random background music during training.
- B. Randomly perturbing mass, friction, and latency in simulation so the policy transfers robustly to reality.
- C. Randomly shutting off power to the computer.

Answer Key: 1-B, 2-A, 3-B, 4-A, 5-B (1-B: 60 floats; 2-A: Energy/jerk suppression; 3-B: Value Coach; 4-A: Hardware protection; 5-B: Robust physics randomization)